IT3091 Machine Learning - Personal Learning Journey

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My work began with the production side of the project and the question behind it: what information could help a grower think about expected output before committing land and resources? I learned that defining the decision and the meaning of each data row matters just as much as choosing a model.

I inspected the production workbook's district, season, crop-category, crop, year, extent, and production fields before cleaning them. For valid rows, yield is calculated as production divided by cultivated extent, giving a target in tonnes per hectare. In my separate crop-yield preprocessing notebook, the source contained 94,755 rows and the prepared dataset retained 60,529 valid records after numeric checks, removal of unusable values, and a crop-yield upper-quantile filter.

That separate workflow taught me to keep a traceable link from the raw workbook to the prepared CSV and saved model outputs. The Random Forest yield model uses district, season, crop category, crop, year, and extent. In the app, a user enters the season, district, year, and hectares; the result is shown as yield per hectare and estimated production for that area.

The saved evaluation used a random 80/20 row split and reported MAE of about 1.99 t/ha, RMSE of about 3.10 t/ha, and R-squared of about 0.676. I learned to describe R-squared as a regression score rather than accuracy, and to explain the split that produced it. I also learned that planned area must be available as an input at the time of use, while the model output remains an estimate rather than a guaranteed harvest.

Working on this stage made reproducibility feel practical: later users need stable field names, units, and saved outputs. If I continued the work, I would test the model on later unseen seasons and include reliable soil, irrigation, or farm-practice information. My contribution helped connect production records to a clear hectares-based yield estimate in AgriPlan.